

Functional Enrichment Analysis (GSEA) of Intestinal IEL and LPL Compartments: Disease-Associated Transcriptional Programs in Crohn's Disease

Bioinformatics Analysis Service 2026-08-05

1. Executive Summary

This report describes the functional enrichment analysis performed on pseudobulk differential expression results derived from the compartment-specific reclustering of intraepithelial (IEL) and lamina propria (LPL) lymphocytes in Crohn's disease (CD) versus control intestinal tissue.

The objective was to characterize disease-associated transcriptional programs at the pathway level, using Gene Set Enrichment Analysis (GSEA) as the primary analytical framework, and to determine whether the resulting biological signatures are consistent with independent published findings on the same tissue compartments.

The analytical workflow included:

- pre-ranked GSEA per cell type, using the DESeq2 Wald statistic as ranking metric
- functional interrogation of four curated gene set collections: MSigDB Hallmark, Reactome, GO Biological Process (semantically simplified), and MSigDB C7 ImmunoSigDB
- cross-referencing of enrichment results against Jaeger et al. (2021, *Nature Communications*), the primary literature source for this tissue system
- identification of cross-cell-type and cross-compartment transcriptional patterns
- explicit documentation of methodological limitations and database artifacts encountered during analysis

Five cell populations met the programmatic QC criteria required for a statistically defensible pseudobulk DESeq2 comparison (≥ 2 donors per condition, ≥ 10 cells per pseudo-sample) and were carried forward to enrichment analysis: three in the IEL

compartment (IEL-Cytotoxic-TRM-like, IEL-TH17, IEL-Cycling-T-cells) and two in the LPL compartment (LPL-Cytotoxic-TRM-like, LPL-Treg).

Enrichment results across all five cell types converge on a small number of transcriptional programs consistently associated with CD, most notably an oxidative phosphorylation signature detected in every cell type and database examined. Several findings independently reproduce disease associations reported by Jaeger et al. (2021), strengthening confidence in the biological validity of the upstream clustering and annotation, despite a known methodological limitation affecting compartment-specific reclustering (see Section 9).

2. Analytical Rationale

2.1 Why GSEA as the primary method

Pseudobulk differential expression in this dataset is constrained by a small number of donors per condition per cell type (as low as 2-4 in several groups). Under these conditions, over-representation analysis (ORA) is fragile: the hypergeometric test depends on an arbitrary significance threshold that discards genes with real but sub-threshold effects, and with small input gene lists a single gene gained or lost can flip a pathway's significance. Pre-ranked GSEA, by contrast, uses the full ranked gene list and does not depend on a hard cutoff, making it substantially more robust to the statistical power constraints inherent to this dataset. GSEA was therefore adopted as the sole enrichment method for this analysis; ORA was evaluated during method development but not included in the final pipeline for the reasons above.

2.2 Ranking metric

Genes were ranked by the DESeq2 Wald statistic (`stat`) rather than by shrunk log2 fold change. The Wald statistic jointly reflects effect size and estimation uncertainty, which is particularly important with small per-condition sample sizes, where fold-change estimates for lowly-expressed genes can be unstable despite shrinkage.

2.3 Gene set collections

Collection	Rationale
MSigDB Hallmark (H)	50 non-redundant, curated gene sets; low-noise first-pass characterization of major biological themes
Reactome	Manually curated, hierarchically organized, higher pathway-level resolution for immune signaling
GO Biological Process	Broadest coverage; used only after semantic redundancy reduction (<code>clusterProfiler::simplify</code> , similarity cutoff = 0.5) to avoid reporting near-duplicate parent/child terms
MSigDB C7 (ImmunoSigDB)	Immune cell-state-specific signatures; most directly relevant to a mucosal immunology dataset, though noisier than the other three collections (Section 9.1)

KEGG was evaluated but excluded from the final analysis: its gene sets are less immune-focused and less frequently updated than Reactome, and non-academic use of KEGG pathway data requires a commercial license, which is a relevant consideration given the intended use of this project.

2.4 Statistical parameters

`pvalueCutoff = 0.05` was used in all `clusterProfiler` / `fgsea` calls to retain only statistically significant enrichment results during the analysis and downstream visualization. GO (BP) enrichment additionally required `eps = 1e-4` and constrained gene set size (`minGSSize = 15` , `maxGSSize = 300`) to make computation tractable; the default `eps = 1e-10` was found to make the ~7,000+ GO (BP) term space computationally intractable on standard hardware (Section 9.3).

3. Eligible Cell Types

Enrichment analysis was restricted to cell types that passed the same programmatic QC gate applied upstream for pseudobulk DESeq2 (Section: *Pseudobulk Differential Expression Analysis* report), requiring at least 2 donors per condition and at least 10 cells per pseudo-sample.

IEL compartment

Eligible cell types: IEL-Cycling-T-cells, IEL-Cytotoxic-TRM-like, IEL-TH17

LPL compartment

Eligible cell types: LPL-Cytotoxic-TRM-like, LPL-Treg

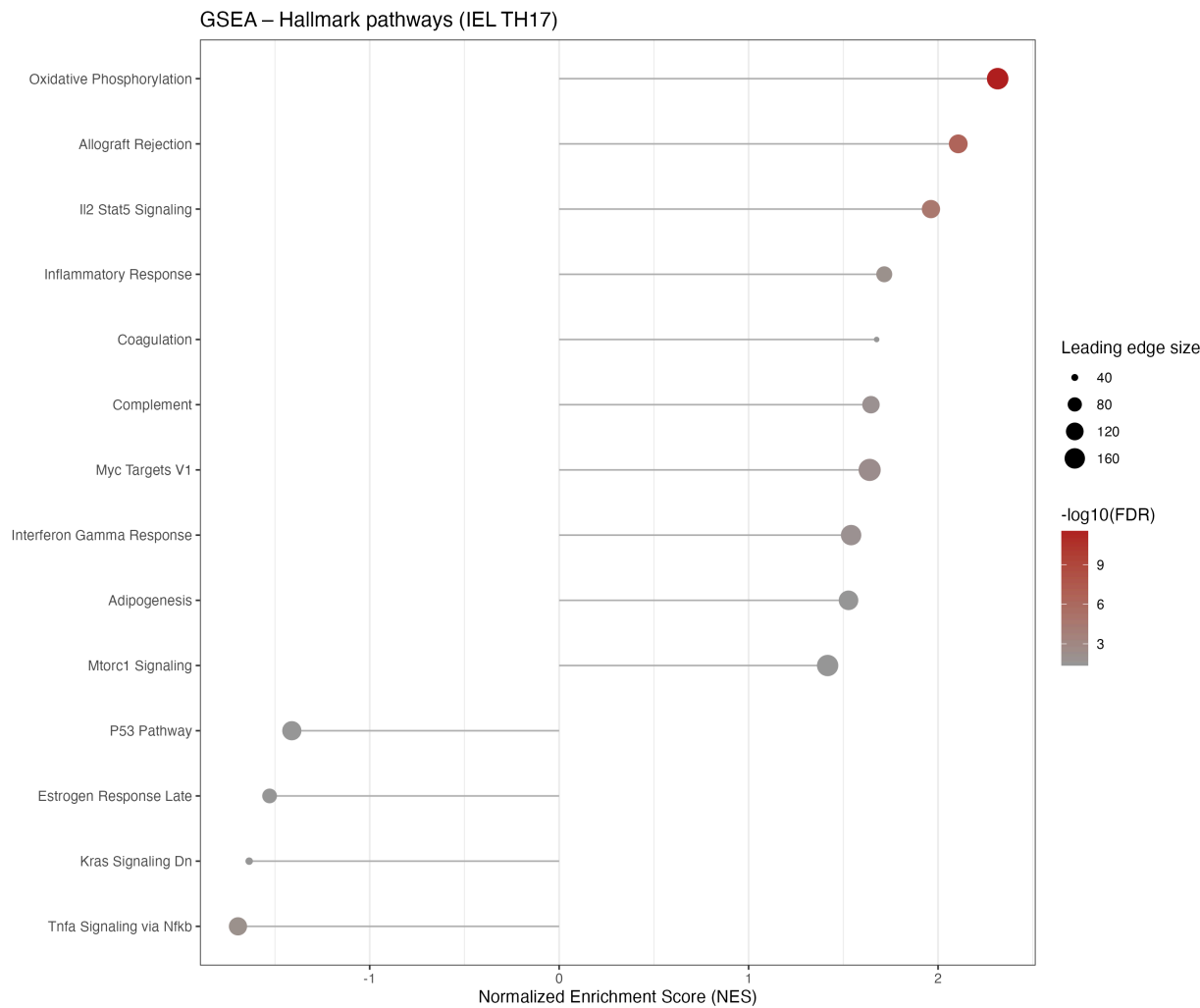
Cell types excluded by this gate (e.g. IEL-TFH, IEL-Treg, IEL-Other-T-cells at finer resolution) reflect genuine structural constraints of the donor cohort rather than an arbitrary analytical choice, and are not analyzed further in this report.

4. IEL Compartment: Enrichment Results

4.1 IEL-TH17

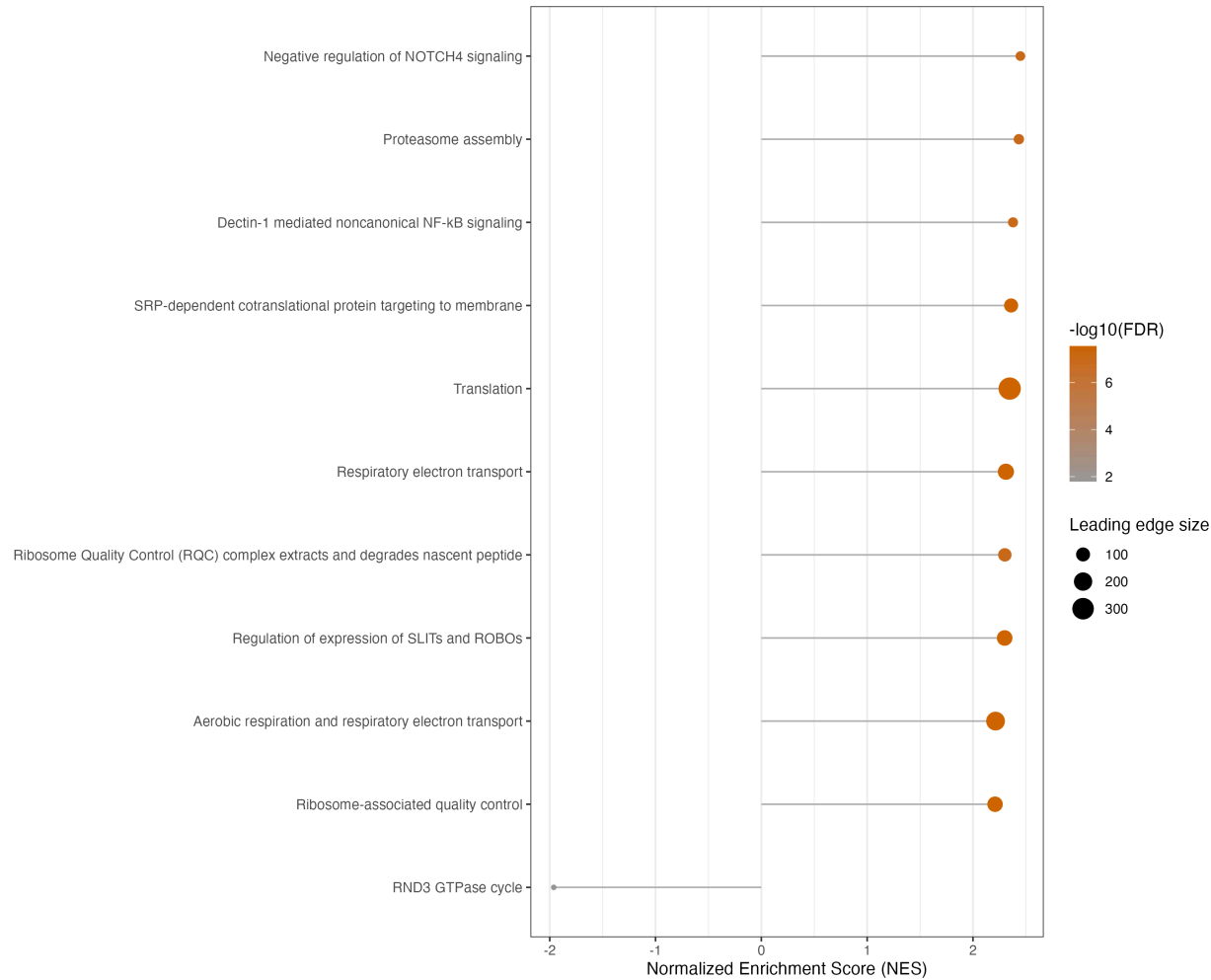
TH17-like CD4 T cells show the most internally coherent enrichment profile in the IEL compartment, with strong agreement across all four databases on an activated, inflammatory phenotype in CD.

Hallmark



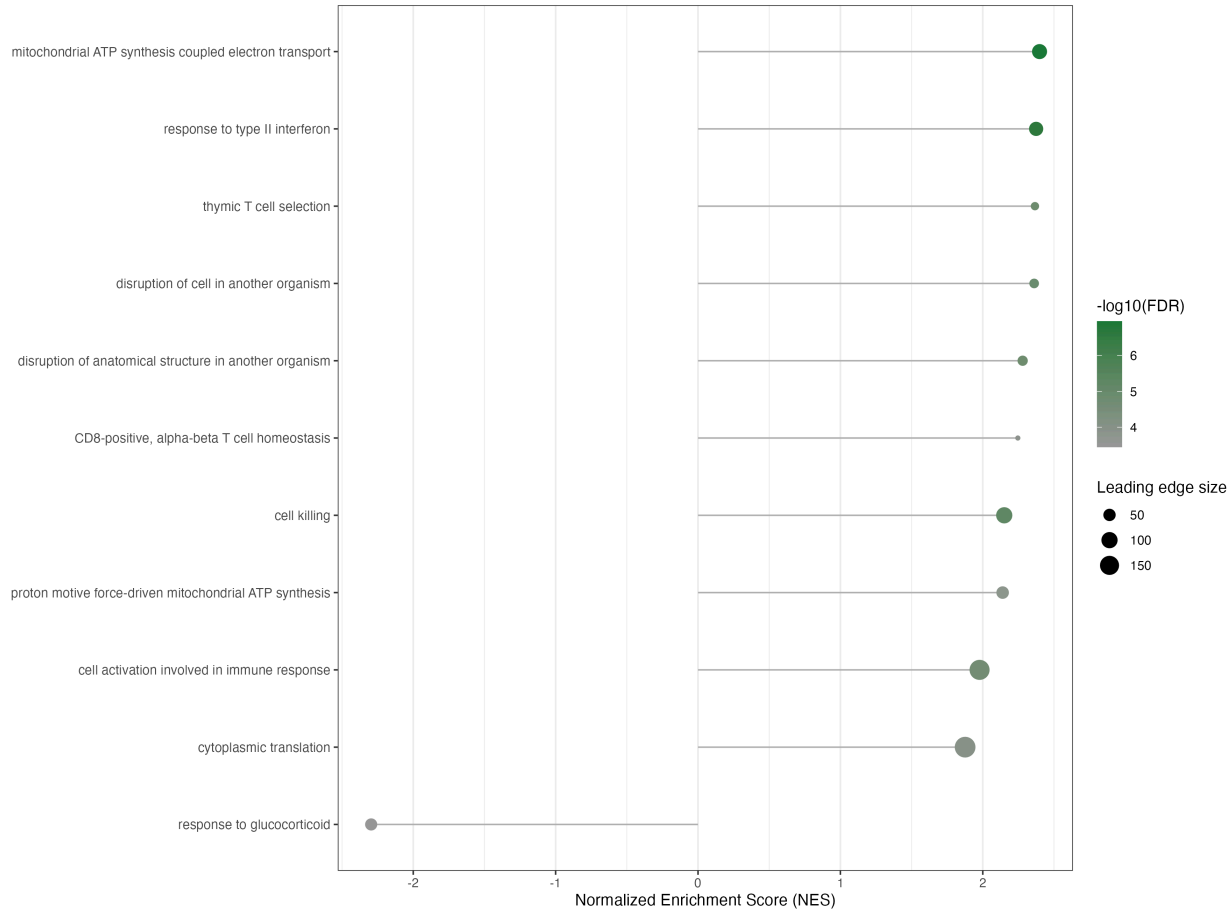
Reactome

REPORT_Functional_enrichment.html
GSEA – Reactome pathways (IEL TH17)

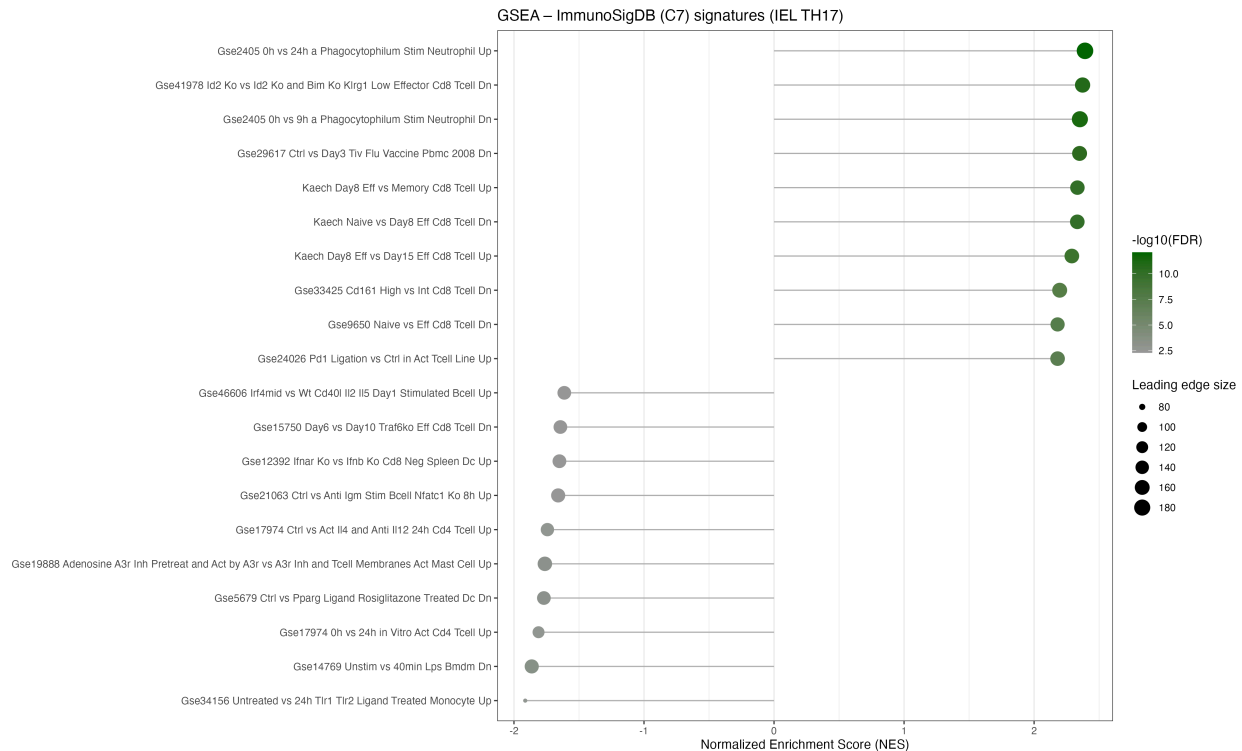


GO Biological Process

GSEA – GO Biological Process (IEL TH17)



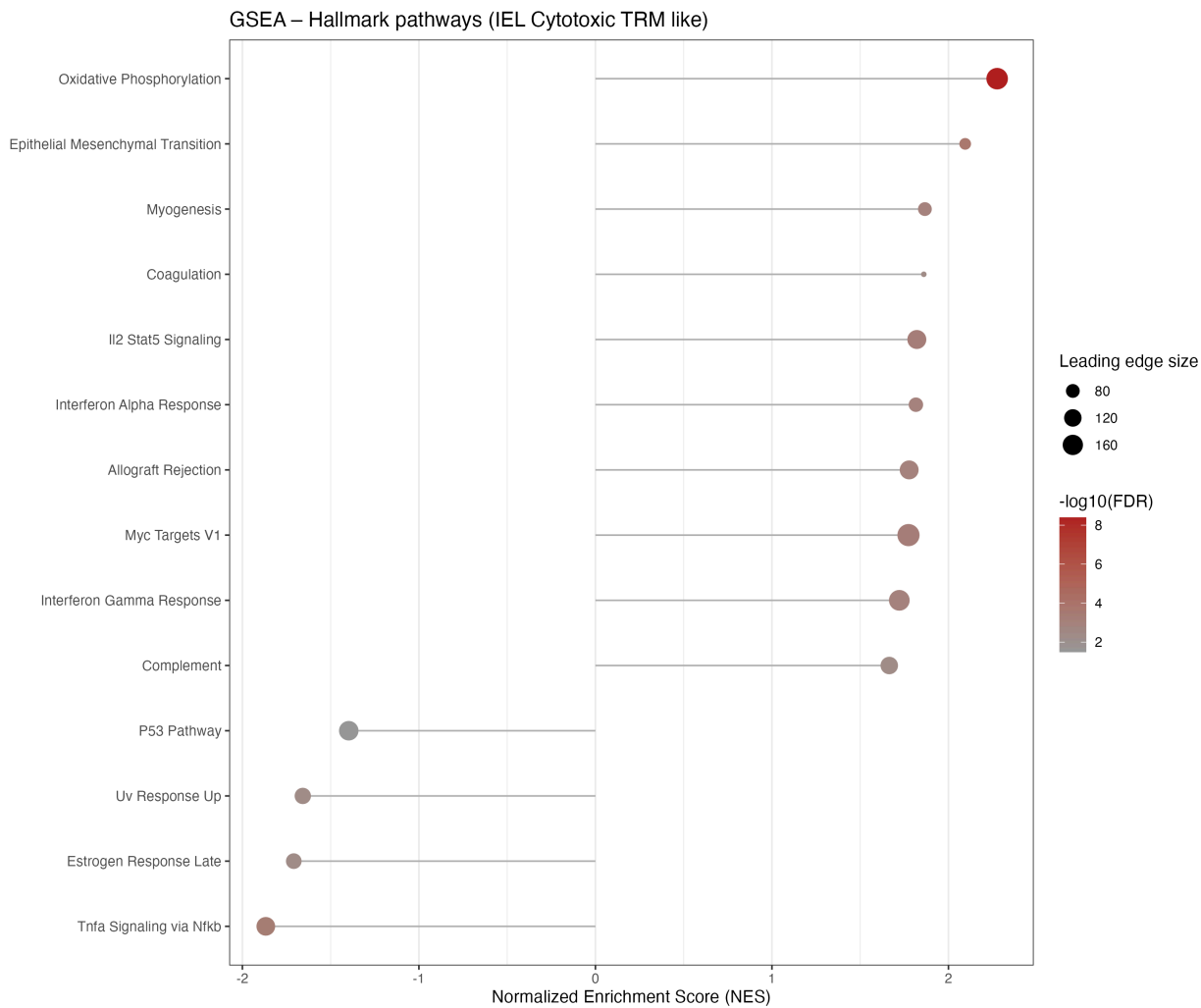
C7 ImmunoSigDB



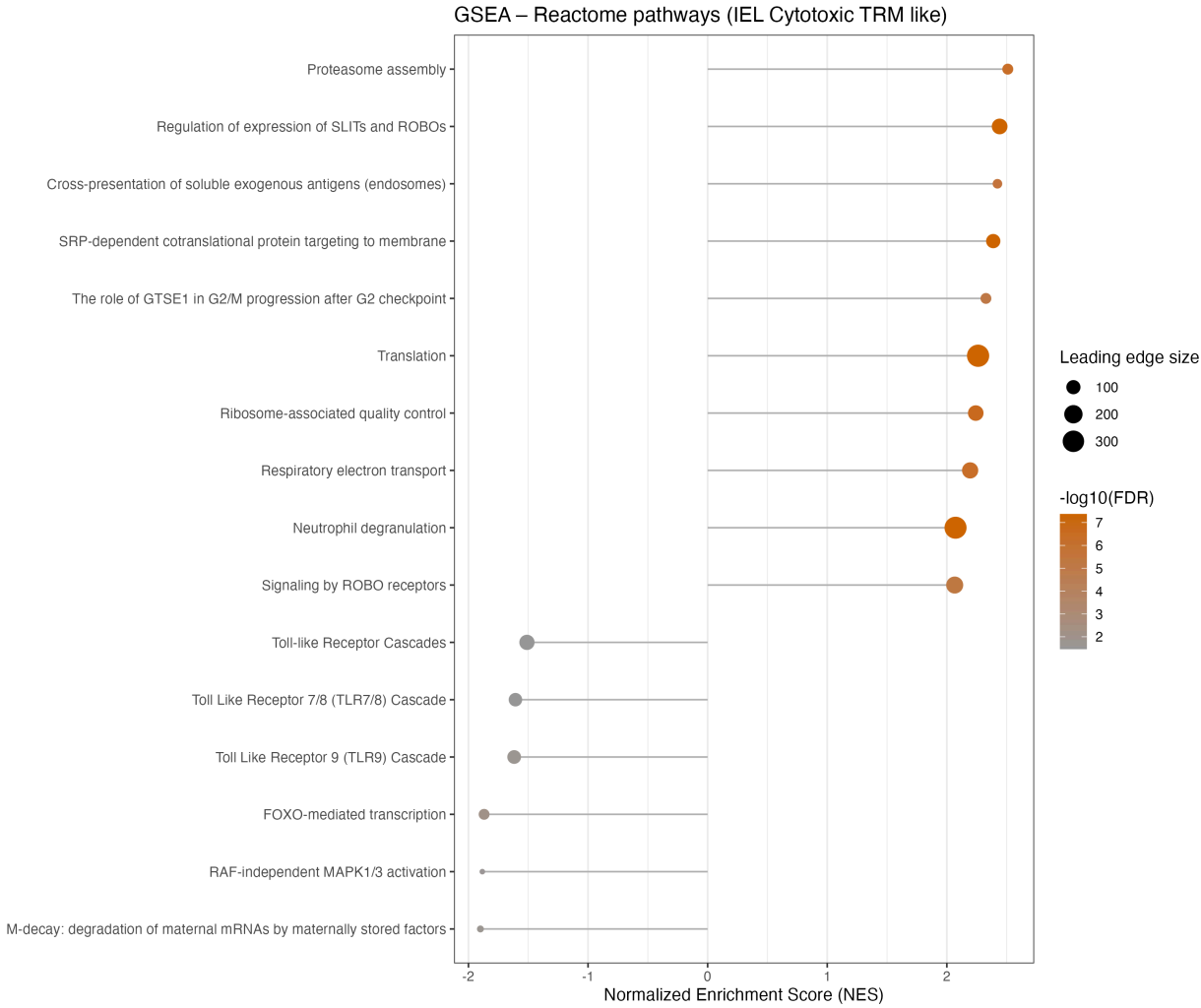
Interpretation. Hallmark shows Inflammatory Response, Interferon Gamma Response, Allograft Rejection, Complement, and IL2-STAT5 Signaling all up in CD — consistent with an activated, cytokine-driven TH17 phenotype. This matches Jaeger et al. (2021), who report that IEL CD39⁺ TH17 cells exhibit pathogenic effector features (GZMB, CCL4 expression) and are increased in CD. Reactome shows a notable divergence between NF- κ B signaling branches: the non-canonical (NIK→noncanonical NF- κ B signaling) branch is up in CD, while Hallmark TNFA Signaling via NF- κ B is down — mechanistically distinct branches (non-canonical driven by CD40/BAFFR/LT β R versus canonical TNFR1/IL1R signaling), and consistent with the CD40-CD40L costimulatory axis reported to sustain TH17 differentiation in this tissue. GO (BP) additionally identifies CD8-positive, alpha-beta T cell homeostasis as enriched, alongside C7 signatures dominated by CD8 effector differentiation comparisons (Kaech Naive vs Effector); manual verification of the underlying cluster composition indicated this population (Seurat clusters 4, 5, 11) is annotated as CD4/TH17 lineage, with cluster 11 specifically labeled TH17-like innate-associated — a plausible source of shared innate/cytotoxic marker expression independent of true CD8 lineage contamination, though this was not verified at the single-marker level (CD8A , NKG7 , GZMK) and is noted as an open question rather than a resolved finding.

4.2 IEL-Cytotoxic-TRM-like

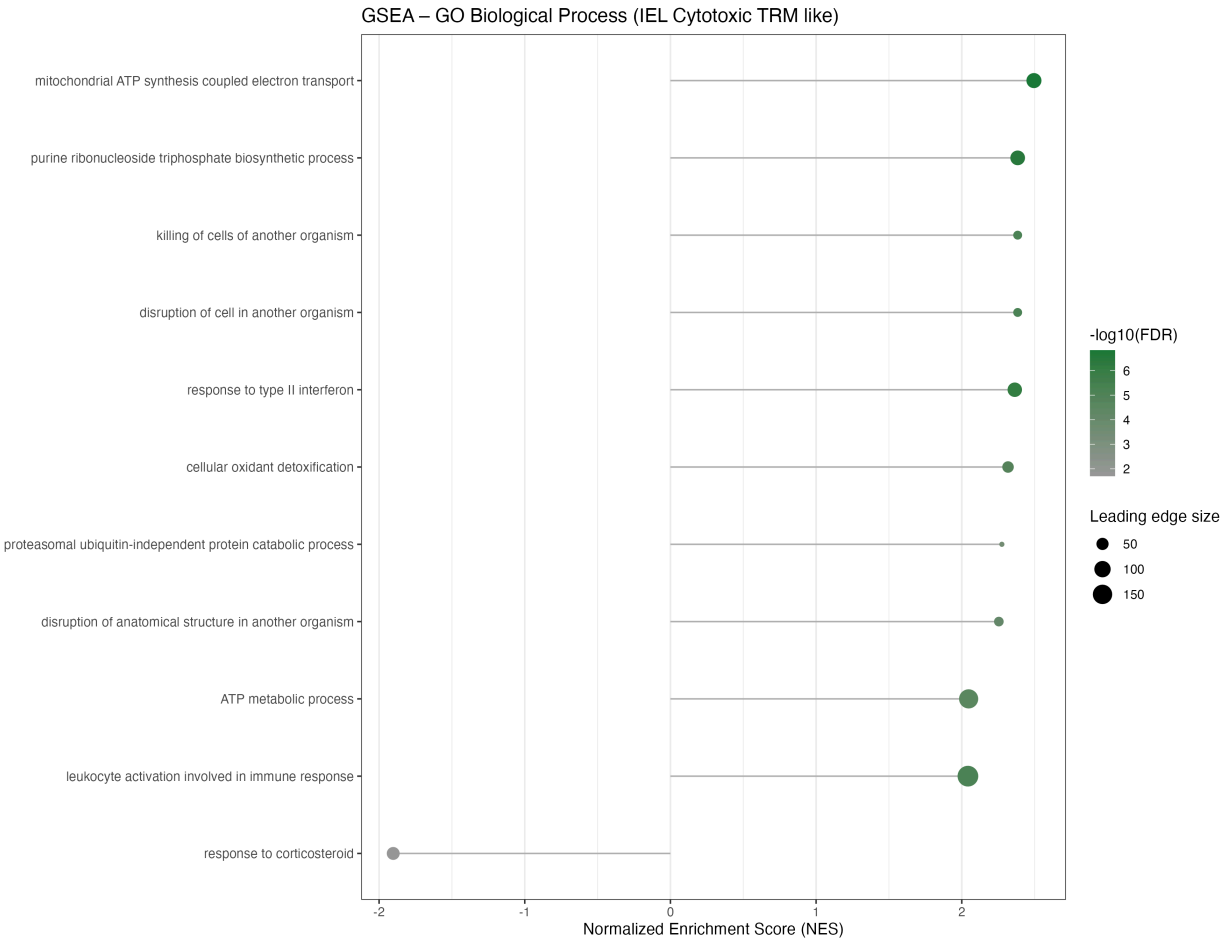
Hallmark



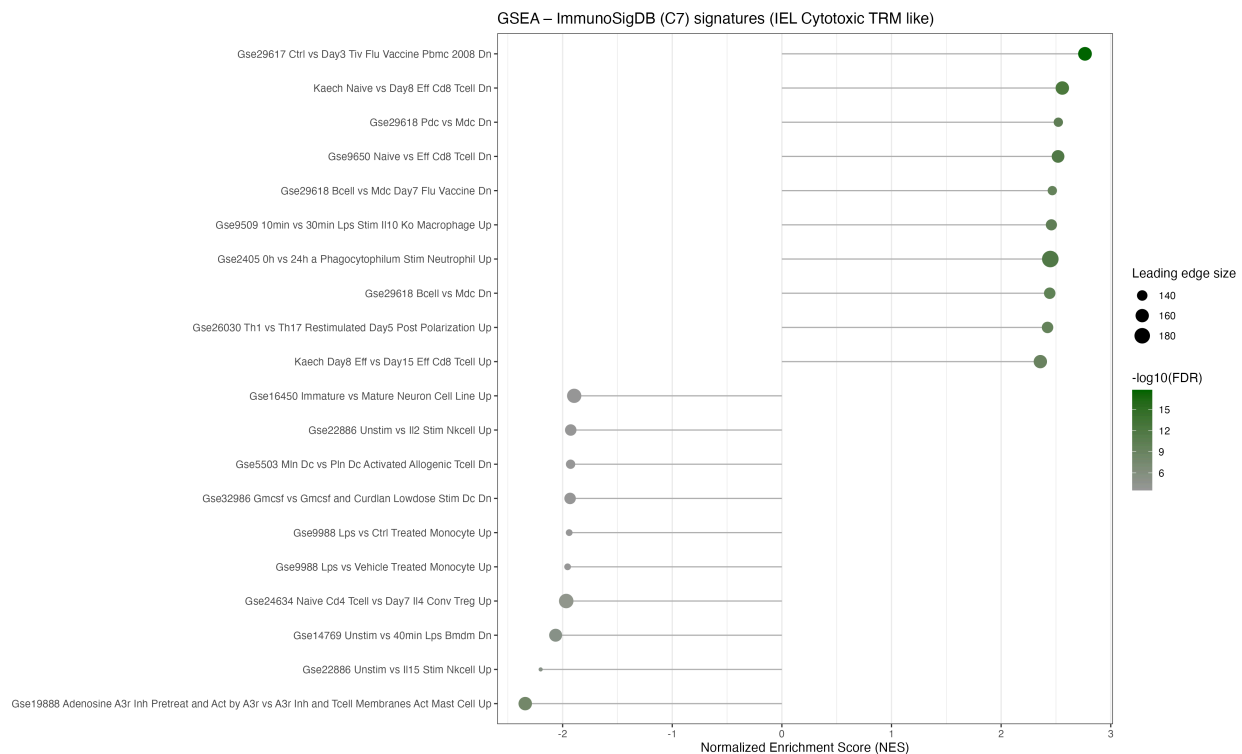
Reactome



GO Biological Process



C7 ImmunoSigDB



Interpretation. This population shows the strongest and most database-consistent oxidative phosphorylation signature in the IEL compartment (Hallmark Oxidative Phosphorylation, GO (BP) mitochondrial ATP synthesis coupled electron transport and five related redundant GO terms collapsing to the same theme after semantic simplification, Reactome Aerobic respiration and respiratory electron transport), alongside Interferon Gamma Response, Allograft Rejection, and Complement, up in CD. C7 signatures are dominated by “Naive vs Effector CD8” comparisons, consistent with a shift toward an early/peak effector CD8 phenotype in CD. GO (BP) additionally identifies response to type II interferon (not type I), distinguishing this population from IEL-Cycling-T-cells (Section 4.3), and response to corticosteroid / glucocorticoid , down in CD — a pattern recurring across three of the five cell types analyzed (Section 7.3).

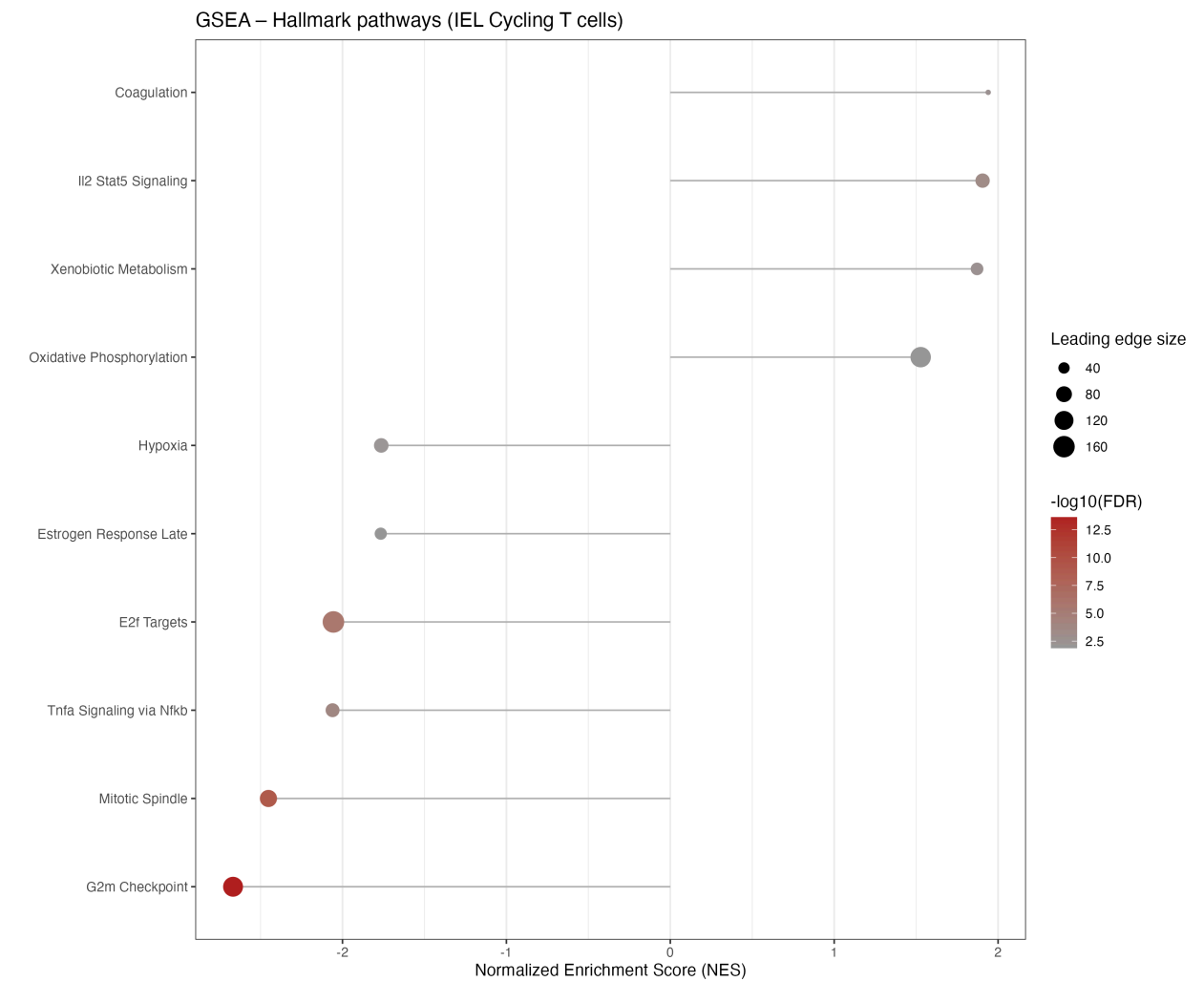
Compositional caveat (see also Section 9.2). This macropopulation aggregates two distinct T cell lineages: CD8 $\alpha\beta$ tissue-resident effector cells (Seurat clusters 0, 10, 14, 15) and cytotoxic $\gamma\delta$ T cells (clusters 1, 2). Separating these into independent pseudobulk units was tested but failed QC (the $\gamma\delta$ subgroup was represented in only one donor). Composition analysis showed that the relative proportion of $\gamma\delta$ cells within this macropopulation drops from 76.2% in Control to 37.6% in CD (Table 1), a shift consistent in direction with the global reduction in IEL $\gamma\delta$ T cells reported in CD by Jaeger et al. (2021). The enrichment results above should therefore be interpreted with the possibility that part of the observed signal reflects compositional shift (relative CD8: $\gamma\delta$ proportions) rather than purely condition-driven transcriptional reprogramming within a fixed population.

Table 1. IEL-Cytotoxic-TRM-like composition by condition

Subpopulation	CD (n cells, %)	Control (n cells, %)
CD8 αβ TRM/effector	3,512 (62.4%)	716 (23.8%)
γδ cytotoxic	2,114 (37.6%)	2,298 (76.2%)

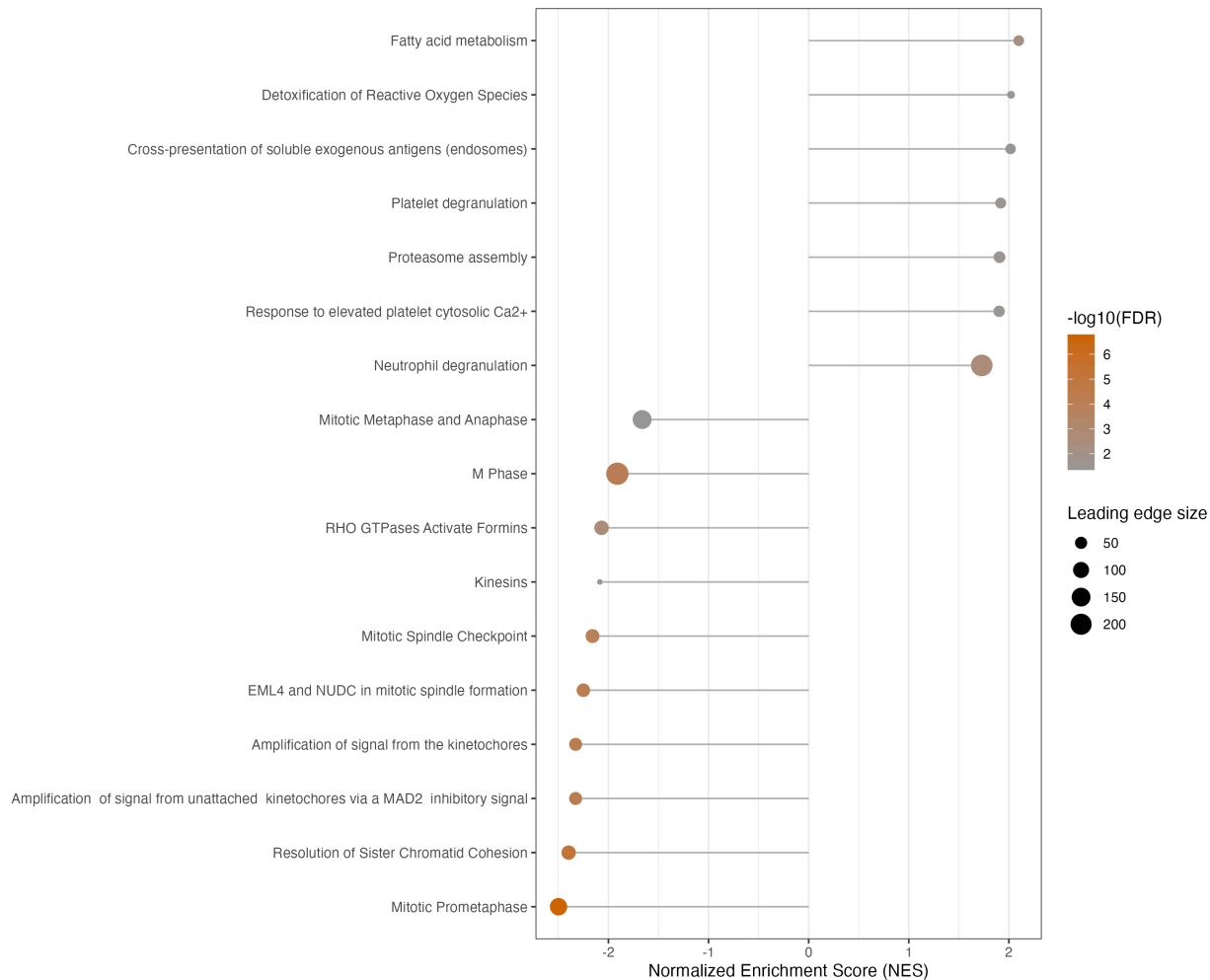
4.3 IEL-Cycling-T-cells

Hallmark



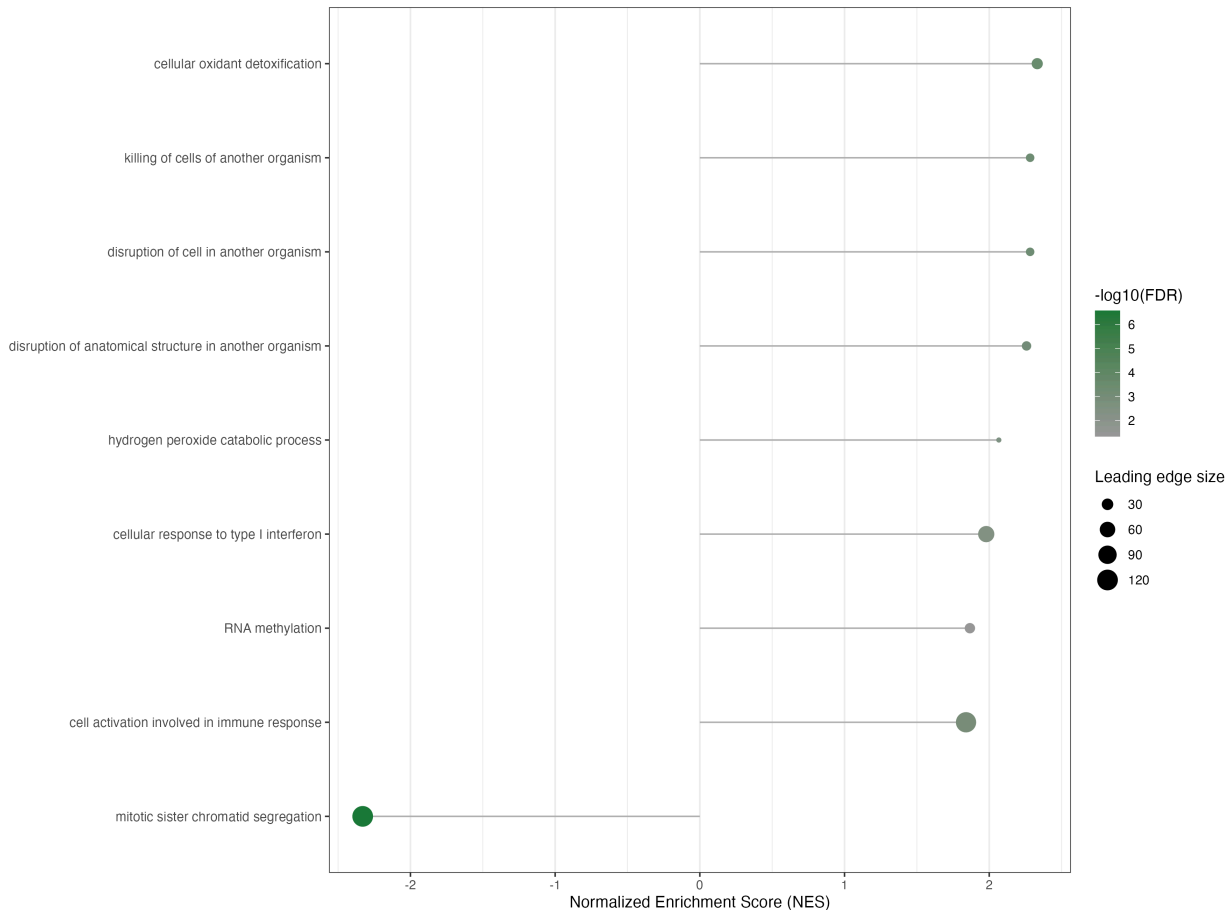
Reactome

GSEA – Reactome pathways (IEL Cycling T cells)

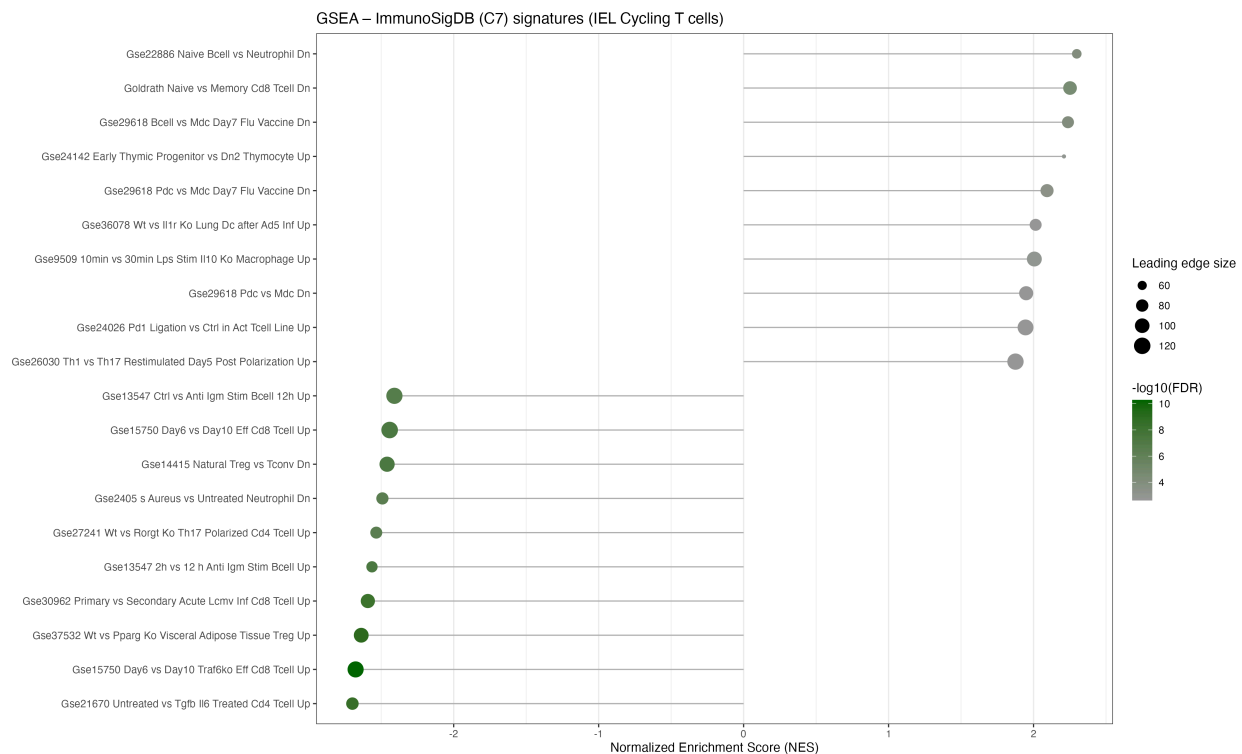


GO Biological Process

GSEA – GO Biological Process (IEL Cycling T cells)



C7 ImmunoSigDB

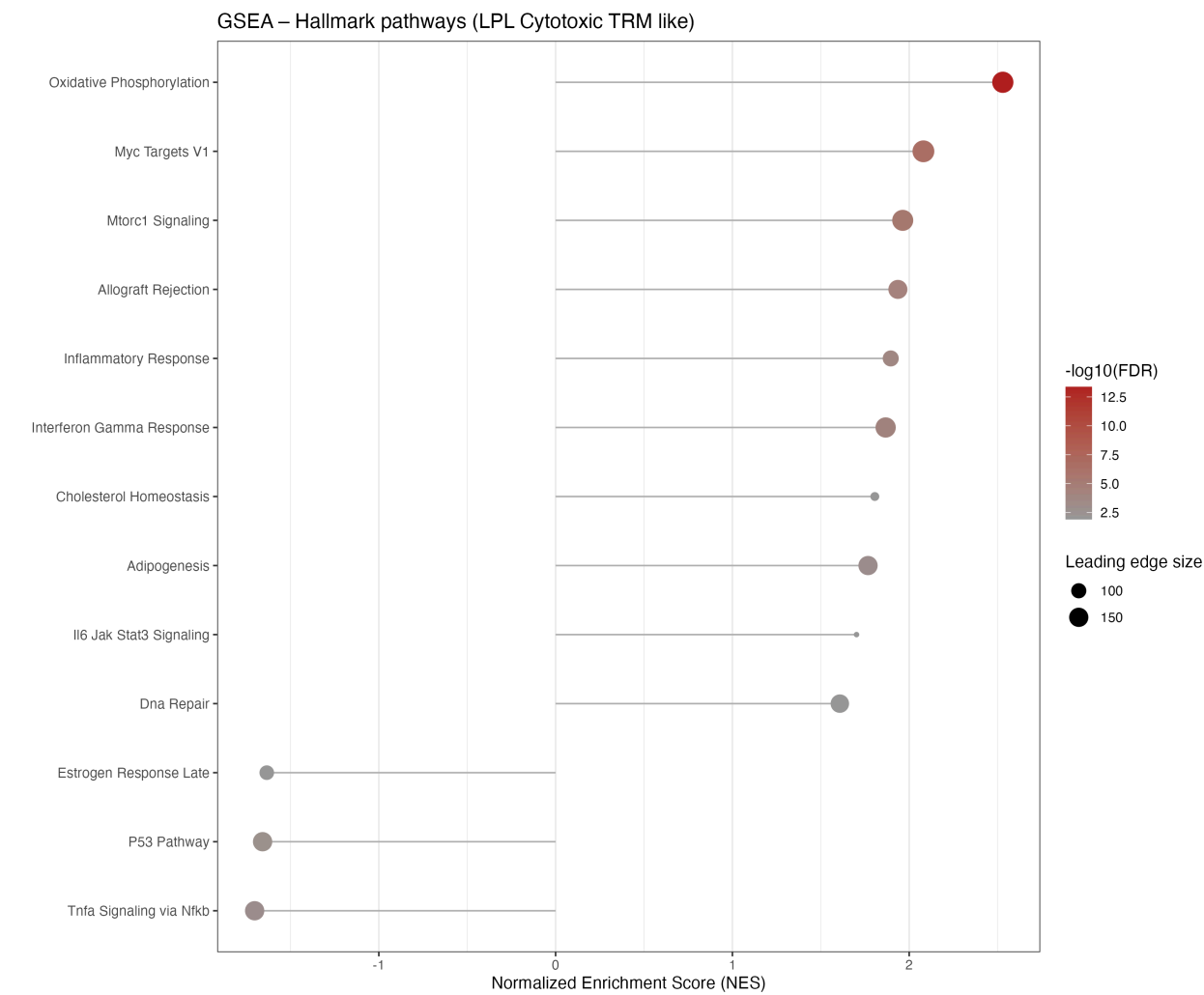


Interpretation. Three independent databases (Hallmark: G2M Checkpoint, E2F Targets, Mitotic Spindle; Reactome: M Phase, Mitotic Prometaphase, Resolution of Sister Chromatid Cohesion; GO (BP): mitotic nuclear division and related chromosome segregation terms, collapsed by semantic simplification) concordantly show a reduced mitotic machinery signature in CD within this already-cycling population, with a parallel increase in oxidative phosphorylation and fatty acid metabolism (Hallmark, Reactome). This convergence across independent gene set collections elevates this observation from speculative to well-supported, although it does not correspond to a comparison explicitly tested in Jaeger et al. (2021) and is presented as a novel, hypothesis-generating finding of this dataset. GO (BP) additionally identifies an antimicrobial/detoxification axis up in CD (killing of cells of another organism, disruption of cell in another organism, cellular oxidant detoxification), together with response to type I interferon — distinguishing this population's interferon signature from the type II response seen in IEL-Cytotoxic-TRM-like and IEL-TH17 (Section 7.4). This antimicrobial signature is consistent with the antibacterial properties reported for S100A family members and IL-26 in this tissue system.

5. LPL Compartment: Enrichment Results

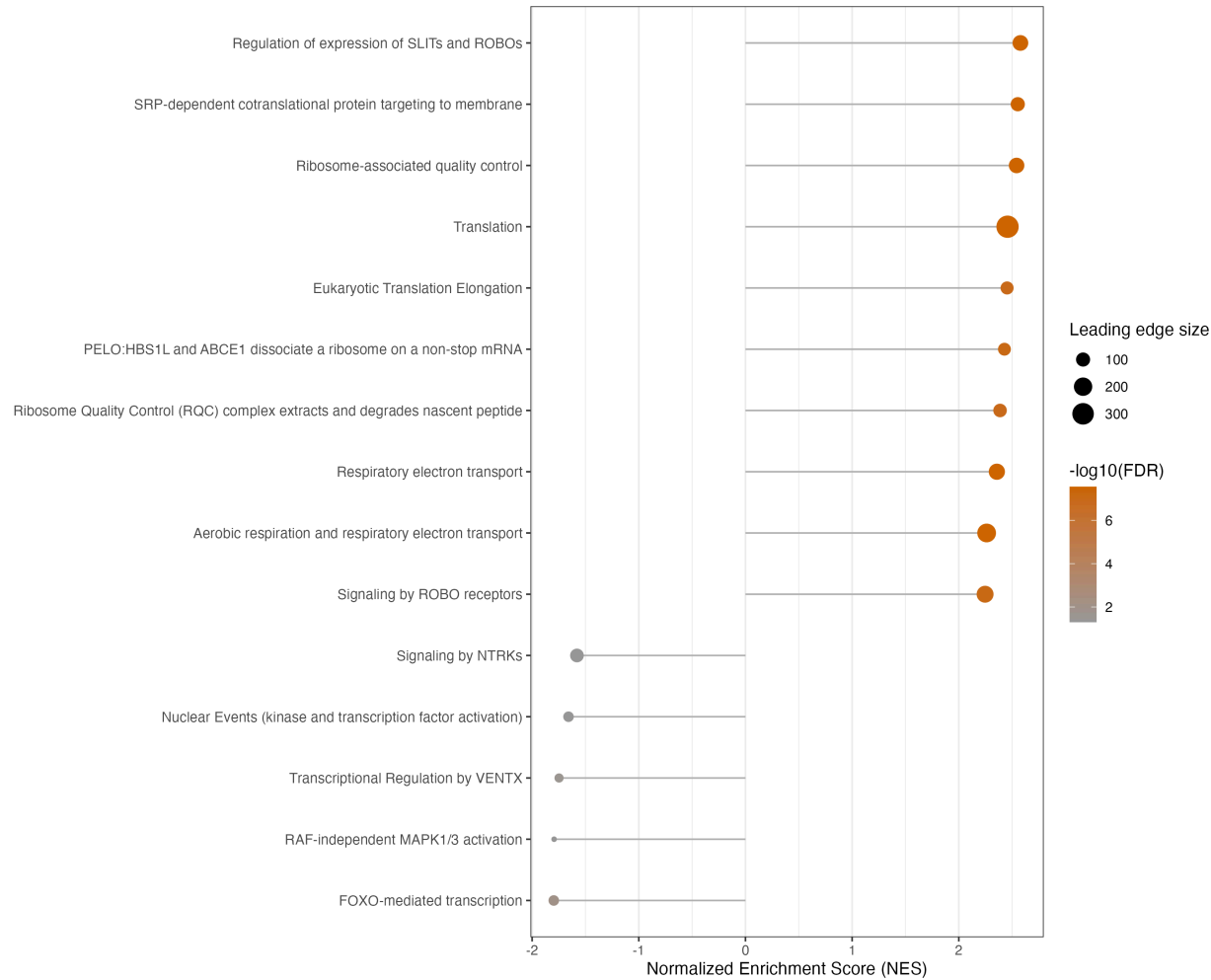
5.1 LPL-Cytotoxic-TRM-like

Hallmark

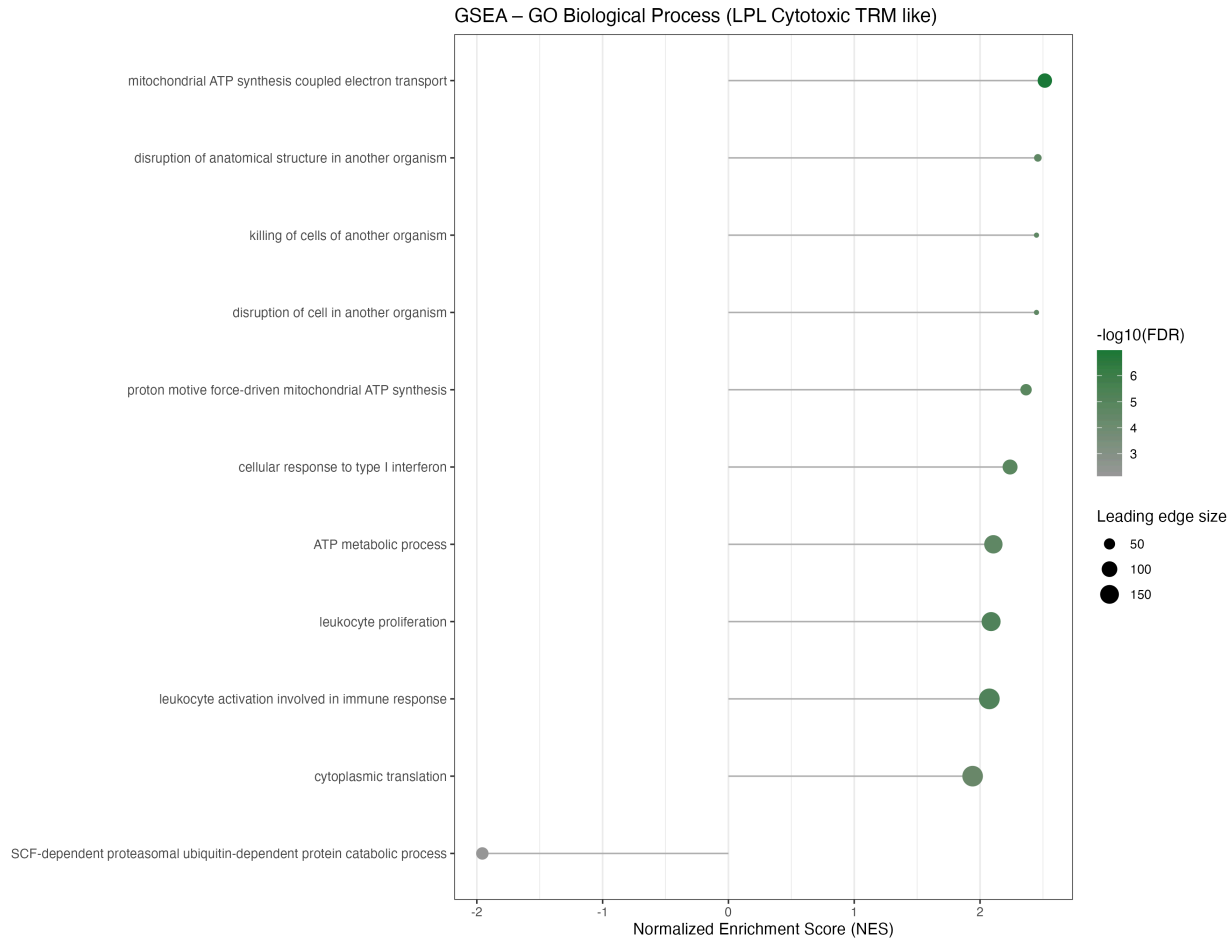


Reactome

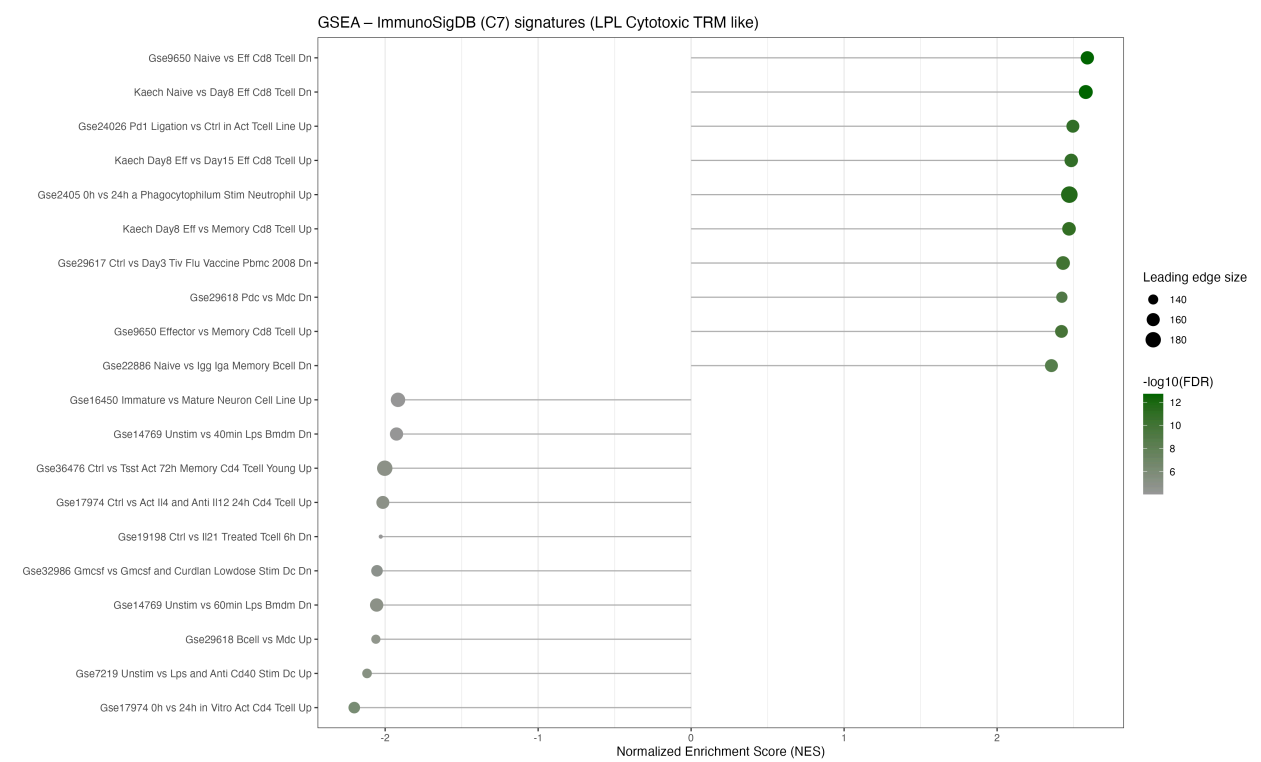
REPORT_Functional_enrichment.html
GSEA – Reactome pathways (LPL Cytotoxic TRM like)



GO Biological Process



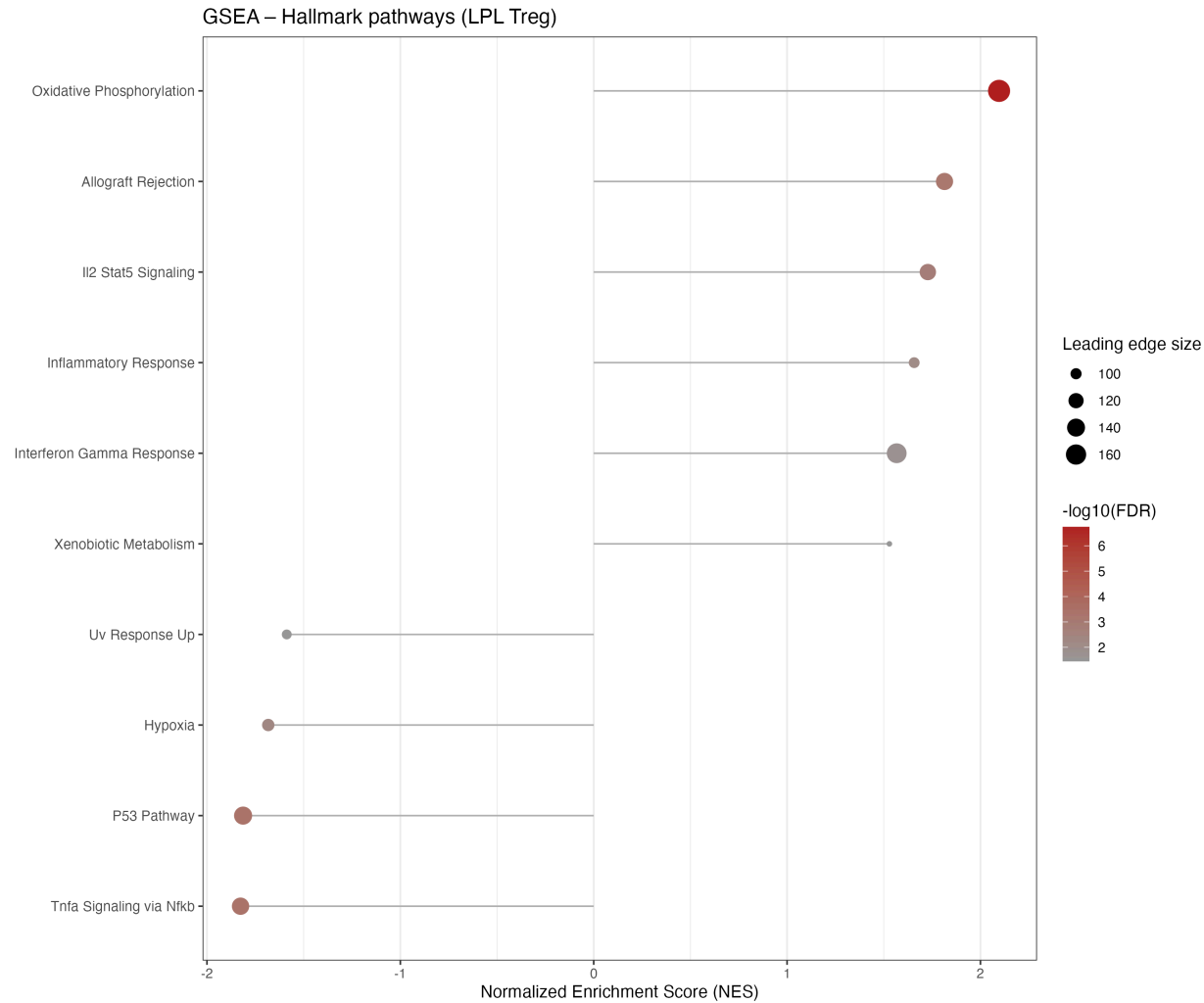
C7 ImmunoSigDB



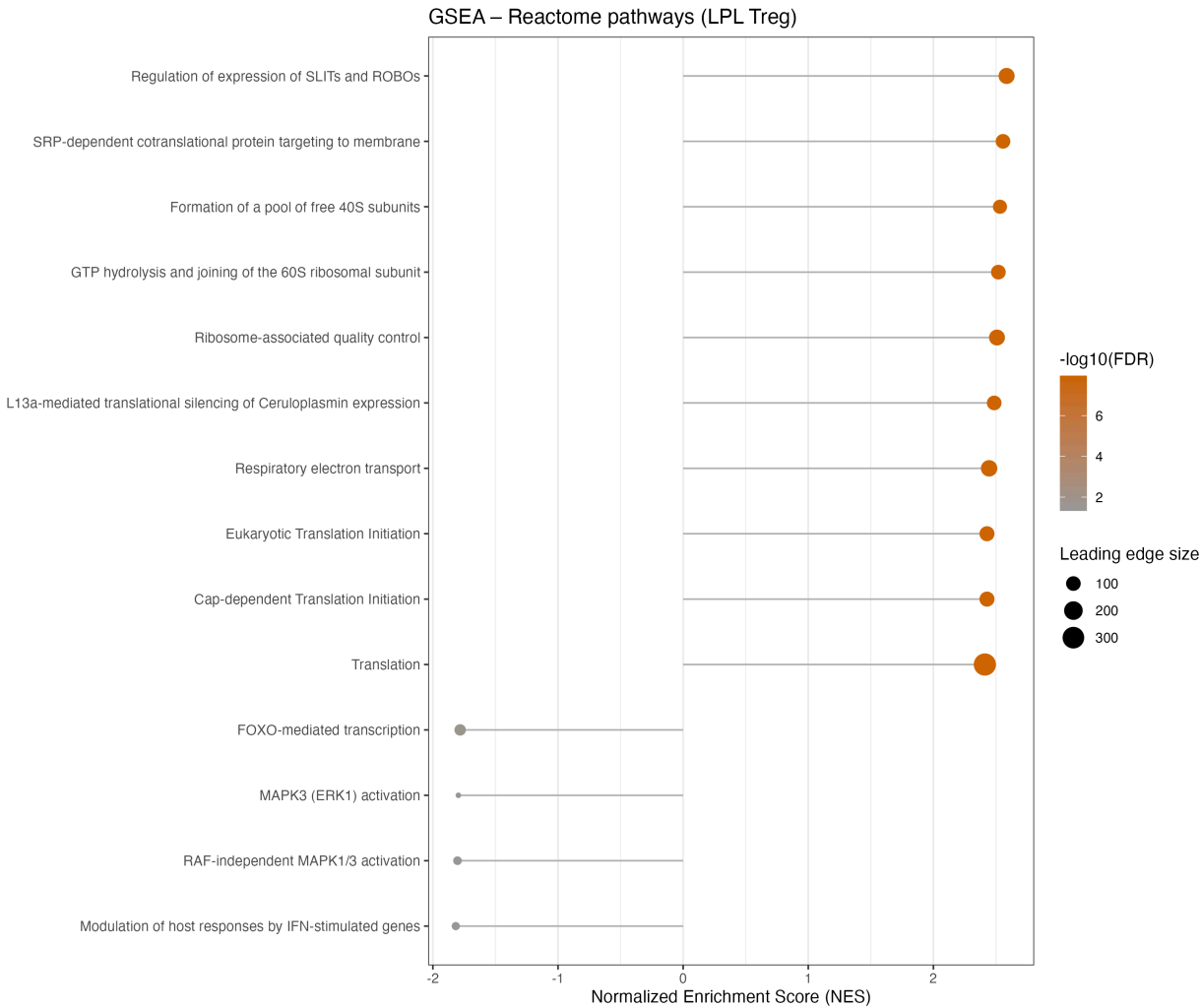
Interpretation. In contrast to IEL, where CD8⁺ IEL T cells are globally reduced in CD, Jaeger et al. (2021) report an *increase* in LP CD8⁺ resident T cells in CD (CyTOF-based expansion of multiple CD8⁺ clusters). The Hallmark profile obtained here is consistent with this opposite compartmental trend: alongside the oxidative phosphorylation signature shared across all cell types in this analysis, this population additionally shows Myc Targets V1 and mTORC1 Signaling up in CD — the canonical transcriptional axes of active clonal expansion, not observed with comparable prominence in any IEL population. Reactome independently supports this interpretation with a strong ribosome biogenesis/translation program (Translation, SRP-dependent protein targeting, Ribosome-associated quality control, Eukaryotic Translation Elongation) up in CD — the same biological process described by two independent databases, consistent with active proliferative/effector expansion rather than a static metabolic shift alone. GO (BP) shows cellular response to type I interferon , distinguishing this compartment’s dominant interferon axis from the type II response more typical of IEL populations (Section 7.4).

5.2 LPL-Treg

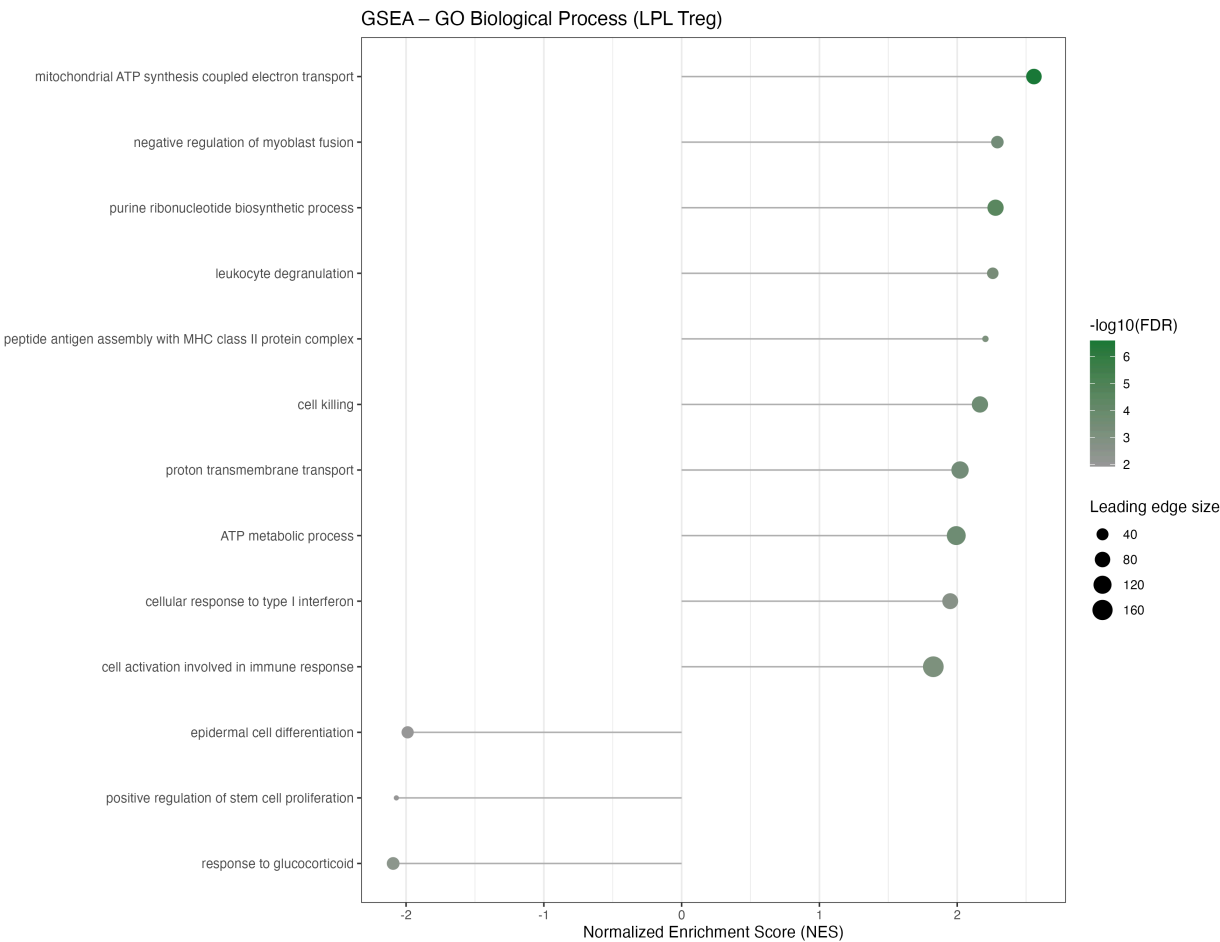
Hallmark



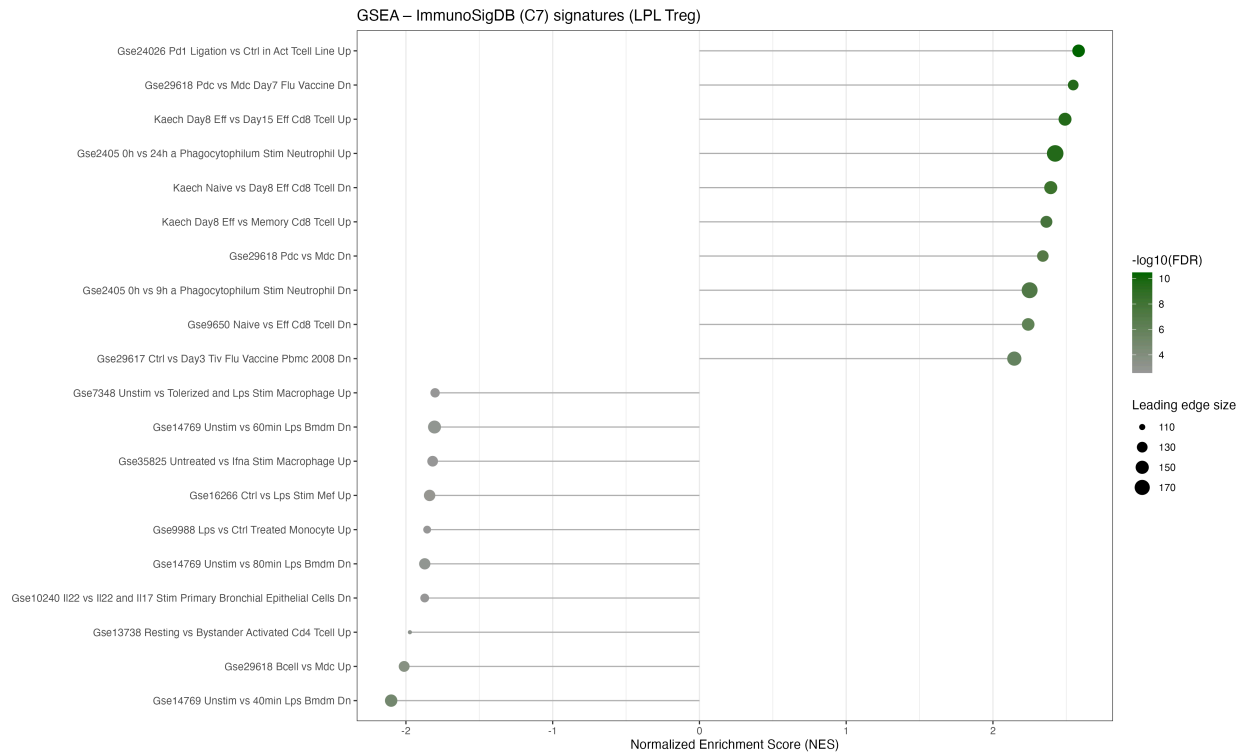
Reactome



GO Biological Process



C7 ImmunoSigDB



Interpretation. The most mechanistically specific finding in this report: Hallmark IL2-STAT5 Signaling is up in CD within the Treg population. IL-2/STAT5 signaling is the canonical molecular axis maintaining Treg identity and function (FOXP3 expression itself is STAT5-dependent), making this a targeted, interpretable signal rather than a generic activation readout. Given that Jaeger et al. (2021) report a reduced proportion of Treg cells in CD lamina propria, this finding is consistent with residual Treg cells being those most actively engaged via IL2-STAT5 signaling — either as an active but insufficient compensatory regulatory response, or reflecting a biased survival of the subset in which this axis is most engaged. GO (BP) additionally identifies peptide antigen assembly with MHC class II protein complex up in CD, consistent with an antigen-presentation-mediated tolerance mechanism distinct from purely cytokine-based suppression. Reactome repeats the same translation/ribosome biogenesis program observed in LPL-Cytotoxic-TRM-like (Section 5.1), indicating this is a compartment-level rather than cell-type-specific pattern distinguishing LPL from IEL (Section 7.2).

6. Comparison with Jaeger et al. (2021)

Finding in this analysis	Jaeger et al. (2021)	Concordance
IEL-TH17: inflammatory/IFN- γ activation up in CD	CD39 ⁺ IEL TH17 increased, pathogenic effector features (GZMB, CCL4) in CD	Direct
IEL-Cytotoxic-TRM-like: $\gamma\delta$ proportion drops 76.2%→37.6% (CD)	Global reduction of IEL $\gamma\delta$ T cells in CD-inflamed tissue	Direct (compositional, quantified)
LPL-Cytotoxic-TRM-like: Myc/mTORC1/translation program up in CD	CD8 ⁺ resident T cells expanded (CyTOF) in LP of CD patients	Consistent, opposite-direction trend from IEL as expected
LPL-Treg: IL2-STAT5 signaling up in CD	Reduced Treg proportion in CD LP	Consistent (residual Treg under altered signaling)
IEL-Cycling-T-cells: reduced mitotic signature, antimicrobial/OXPHOS shift	Not directly tested in source paper	Novel, hypothesis-generating

7. Cross-Compartment Biological Synthesis

7.1 A universal oxidative phosphorylation signature

Across all five analyzed cell types, in both compartments and across every database queried, an oxidative phosphorylation / mitochondrial ATP synthesis signature is up in CD. This is the single most robust and reproducible finding of the entire analysis and is proposed as the primary, highest-confidence result of this study.

7.2 Compartment-level divergence in expansion programs

LPL populations (both Cytotoxic-TRM-like and Treg) show a shared translation/ribosome biogenesis program not observed with comparable strength in any IEL population, consistent with the qualitative description of LP as a more actively regulatory, cytokine- and expansion-driven niche relative to the more transcriptionally static, tissue-resident IEL niche (Section 2, Subsetting report).

7.3 A recurring corticosteroid/glucocorticoid signature

Response to corticosteroid / glucocorticoid is down in CD in three of the five cell types analyzed (IEL-Cytotoxic-TRM-like, IEL-TH17, LPL-Treg), spanning both compartments. This pattern has plausible clinical relevance given the known issue of corticosteroid resistance in refractory CD, but is reported here as an observational, hypothesis-generating signal rather than a mechanistically resolved finding.

7.4 Type I versus type II interferon axis

A consistent distinction emerges between an interferon type II (IFN- γ) response (IEL-Cytotoxic-TRM-like, IEL-TH17) and an interferon type I response (IEL-Cycling-T-cells, LPL-Cytotoxic-TRM-like). This fine-grained distinction is noted as an observation warranting further investigation rather than a claim of distinct upstream regulatory mechanisms.

8. Deliverables

- Pre-ranked GSEA result tables (Hallmark, Reactome, GO (BP), C7) per cell type, per compartment
- Ranked gene lists (Entrez-mapped, stat -ordered) per cell type
- Lollipop enrichment plots per cell type per database (this report)
- Session information logs for full computational reproducibility

9. Future Directions and Recommended Next Steps

The findings in this report point to specific, low-cost follow-up analyses that would strengthen or resolve several observations flagged as provisional above, without requiring new sequencing data.

9.1 Leading-edge gene verification

Two recurring signals require direct inspection of their leading-edge gene sets before being reported as confirmed findings:

- **TNFA Signaling via NF-κB** (down in CD, 4/5 cell types): extract the leading-edge genes from the stored `hallmark_list` objects and check whether the signal is driven by negative feedback regulators (`TNFAIP3` , `NFKBIA` , `ZFP36`) versus canonical pro-inflammatory targets (`CXCL8` , `IL6` , `PTGS2`). The two scenarios support materially different biological narratives (chronic desensitization versus genuine pathway suppression).
- **Response to corticosteroid/glucocorticoid** (down in CD, 3/5 cell types): confirm the leading-edge set includes canonical glucocorticoid-responsive genes (`FKBP5` , `TSC22D3`) rather than a coincidental overlap with a differently-regulated process.

9.2 LPL compositional analysis

The compositional confound identified for IEL-Cytotoxic-TRM-like (Section 4.2, Table 1) was not systematically checked for LPL macropopulations. Given that LPL-Cytotoxic-TRM-like and LPL-Treg were each consolidated from multiple Seurat clusters during annotation (Subsetting Compartment report, Section 7.2), an analogous cluster-by-condition composition table is recommended before treating LPL enrichment results as free of compositional shift effects.

9.3 Cross-cell-type NES visualization

A single summary figure — a heatmap of NES values (pathway × cell type, both compartments) restricted to the pathways discussed in Section 7 (oxidative phosphorylation, TNFA/NF-κB, corticosteroid response, type I/II interferon) — would visually consolidate the cross-compartment synthesis currently presented as text only.

10. Limitations

10.1 Database-specific artifacts

C7 (ImmunoSigDB) is the noisiest of the four collections used: being composed of thousands of heterogeneous pairwise comparisons from unrelated experimental systems (e.g. neutrophils stimulated with *A. phagocytophilum*, LPS-treated monocytes), it can capture generic activation/metabolic programs shared across cell types rather than genuine cell-state specificity. Thematically coherent C7 signatures (e.g. Naive vs Effector CD8, Th1 vs Th17) were prioritized in interpretation; thematically unrelated hits were treated with caution and not reported as primary findings.

“Regulation of expression of SLITs and ROBOs” (Reactome) recurred across all four Reactome analyses performed in this study (both compartments, multiple cell types unrelated by lineage or condition direction). Given axon-guidance biology has no plausible relevance to this system, this is flagged as a likely structural artifact of the Reactome gene set (probable overlap with generic hub signaling regulators) rather than a genuine biological signal, and was excluded from primary interpretation throughout this report.

KEGG was excluded from this analysis (Section 2.3); if required for a specific downstream application, note the commercial licensing requirement for non-academic use.

10.2 IEL-Cytotoxic-TRM-like lineage heterogeneity

As detailed in Section 4.2, this macropopulation aggregates CD8 $\alpha\beta$ and $\gamma\delta$ T cell lineages, which could not be statistically separated due to single-donor representation of the $\gamma\delta$ subgroup. Enrichment results for this population should be interpreted with this compositional caveat explicitly in mind.

10.3 Computational considerations

Default GO (BP) GSEA parameters ($\text{eps} = 1\text{e-}10$) were computationally intractable given the ~7,000+ GO (BP) term space combined with limited local computational resources; $\text{eps} = 1\text{e-}4$ combined with constrained gene set size bounds ($\text{minGSSize} = 15$, $\text{maxGSSize} = 300$) was required to obtain results in tractable time without materially affecting downstream biological interpretation, since the precision lost concerns only the exact magnitude of already highly significant p-values, not their qualitative significance.

10.4 Not independently verified

The TNFA Signaling via NF- κ B down-in-CD signal, recurring in four of five cell types, has not been verified at the leading-edge gene level to distinguish negative-feedback regulators (e.g. TNFAIP3 , NFKBIA) from canonical pro-inflammatory targets, and should be treated as provisional pending this check.

11. Reproducibility

Environment:

- R version: 4.6.0
- Seurat version: 5.5.0
- DESeq2 version: 1.52.0
- apegglm version: 1.34.0
- clusterProfiler version: 4.20.0
- fgsea version: 1.38.0
- msigdb version: 26.1.0
- ReactomePA version: 1.56.0
- org.Hs.eg.db version: 3.23.1

All intermediate GSEA result objects are stored under:

`objects/pseudobulk/`

All visual outputs under:

IEL Compartment: `results/plots/GSEA_IEL/`

LPL Compartment: `results/plots/GSEA_LPL/`

12. Conclusion

Functional enrichment analysis of the IEL and LPL compartments identifies a small set of robust, cross-validated transcriptional programs associated with Crohn's disease, most notably a universal oxidative phosphorylation signature and multiple cell-type-specific programs that independently reproduce findings from Jaeger et al. (2021), including TH17 inflammatory activation, IEL $\gamma\delta$ compositional reduction, LP CD8⁺ expansion, and altered Treg IL2-STAT5 signaling. These convergences, obtained across

independent gene set databases and against an independent literature source, constitute meaningful indirect validation of the upstream clustering, annotation, and pseudobulk differential expression framework, notwithstanding the explicitly documented methodological limitations in Sections 10.2 and 10.3.

Appendix A: Glossary of Abbreviations

Term	Definition
CD	Crohn’s disease
IEL	Intraepithelial lymphocytes
LPL	Lamina propria lymphocytes
TRM	Tissue-resident memory (T cell)
GSEA	Gene Set Enrichment Analysis
ORA	Over-representation analysis
NES	Normalized Enrichment Score
FDR	False discovery rate (Benjamini-Hochberg adjusted p-value)
HVG	Highly variable gene
C7 / ImmunoSigDB	MSigDB immunologic signature gene set collection
GO (BP)	Gene Ontology, Biological Process branch
Wald statistic (stat)	DESeq2 test statistic combining log2 fold change and its standard error

Appendix B: File Manifest

Result objects (objects/pseudobulk/)

- GSEA_ranked_lists_IEL.rds

GSEA_hallmark_list_IEL.rds

GSEA_reactome_list_IEL.rds
- GSEA_ranked_lists_LPL.rds

GSEA_hallmark_list_LPL.rds

GSEA_reactome_list_LPL.rds

GSEA_GOBP_list_IEL.rds

GSEA_GOBP_list_LPL.rds

GSEA_C7_list_IEL.rds

GSEA_C7_list_LPL.rds

Result tables (results/tables/GSEA_IEL/ , results/tables/GSEA_LPL/)

GSEA_hallmark_H_<celltype>_<compartment>.csv

GSEA_reactome_<celltype>_<compartment>.csv

GSEA_GOBP_<celltype>_<compartment>.csv

GSEA_C7_<celltype>_<compartment>.csv

Figures (results/plots/GSEA_IEL/ , results/plots/GSEA_LPL/)

GSEA_Hallmark_<celltype>.png

GSEA_Reactome_<celltype>.png

GSEA_GOBP_<celltype>.png

GSEA_C7_<celltype>.png

GSEA_enrichmentscore_top_<celltype>.png

Session logs (results/logs/)

sessionInfo_GSEA_IEL.txt

sessionInfo_GSEA_LPL.txt